

Distribution analysis

TASK -3

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Distribution Analysis: We use different method to analyze score distribution across population segments.

The primary objective of this section is to examine the underlying statistical structure of the mental health (MH) prevalence data from the APMS 2014 dataset.

In this task, we analyze the *Prevalence_Observed_%* variable across all disorder categories. We employ three specific visualization techniques to profile the data:

- **Quantile-Quantile (Q-Q) Plots:** To compare our sample distribution against a theoretical normal distribution.
- **Segmented Boxplots:** To compare the spread and variance of prevalence across Men, Women, and Geographic Regions.
- **Histograms with Kernel Density Estimates (KDE):** To visualize the frequency of different prevalence rates.

In our specific dataset, we cannot analyze the **raw scores** (e.g., a person scoring a 14 on a PHQ-9 scale) because the data is **aggregated**.

However, we can analyze the **Severity Distribution** based on those scales. For many disorders in this dataset, the "Disorder" column actually lists different score thresholds from the scales.

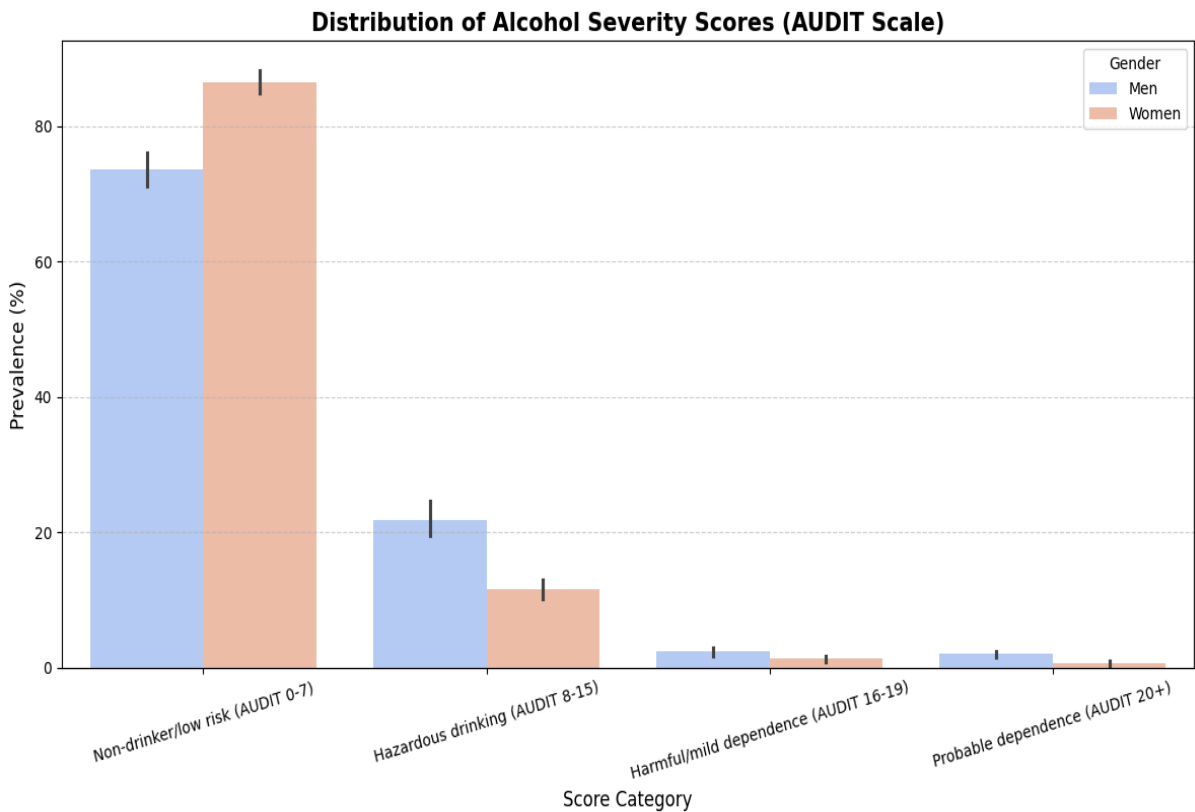
1. Score Distribution using AUDIT Scale:

How we Scale in our dataset : While we don't have individual scores, the researchers categorized the population based on their scores. For example, in the **Alcohol Use** category, the dataset uses the **AUDIT scale** (Alcohol Use Disorders Identification Test):

- **Low Risk:** AUDIT Score 0–7
- **Hazardous Drinking:** AUDIT Score 8–15
- **Harmful/Mild Dependence:** AUDIT Score 16–19
- **Probable Dependence:** AUDIT Score 20+

AUDIT Score Prevalence by Gender:		
Sex	Men	Women
Disorder		
Non-drinker/low risk (AUDIT 0-7)	73.703473	86.510139
Hazardous drinking (AUDIT 8-15)	21.883623	11.612898
Harmful/mild dependence (AUDIT 8+)	NaN	NaN
Harmful/mild dependence (AUDIT 16-19)	2.404989	1.311725
Probable dependence (AUDIT 20+)	2.007915	0.635893
High-risk drinking (AUDIT 16+)	4.412904	1.876963

By analyzing the prevalence of these specific rows, we can essentially analyzing the **distribution of scores** across the population.



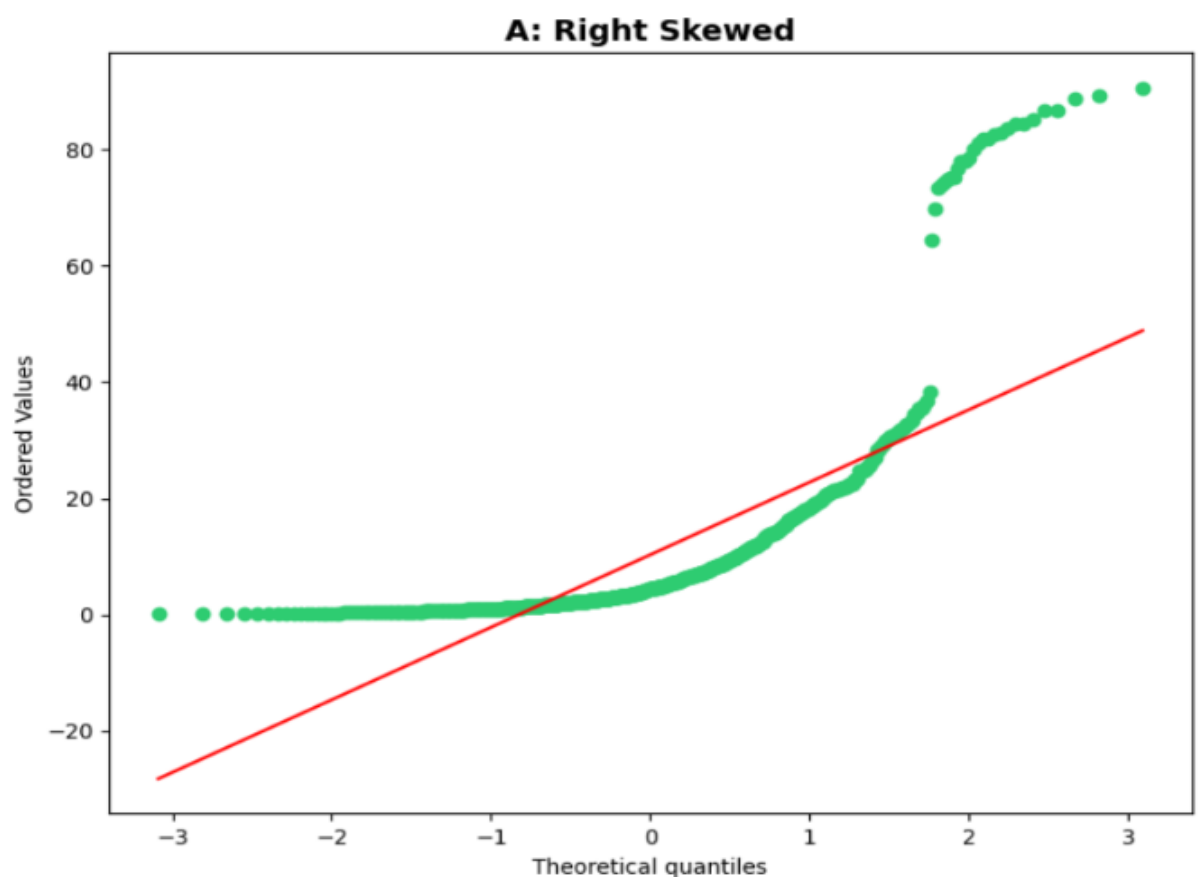
We analyze that:

- **The "Peak" (0–7 scores):** There is vast majority of the population (the highest bars) falls into the 0–7 score range.
- **The "Tail" (16+ scores):** As the "scores" increase, the prevalence drops significantly.
- **Gender Gaps at High Scores:** We can see if men or women are more likely to reach "Severe" scores. In the APMS data, men typically have higher bars in the 8-15 and 20+ categories.

2. Q-Q Plots (quantile-quantile plot):

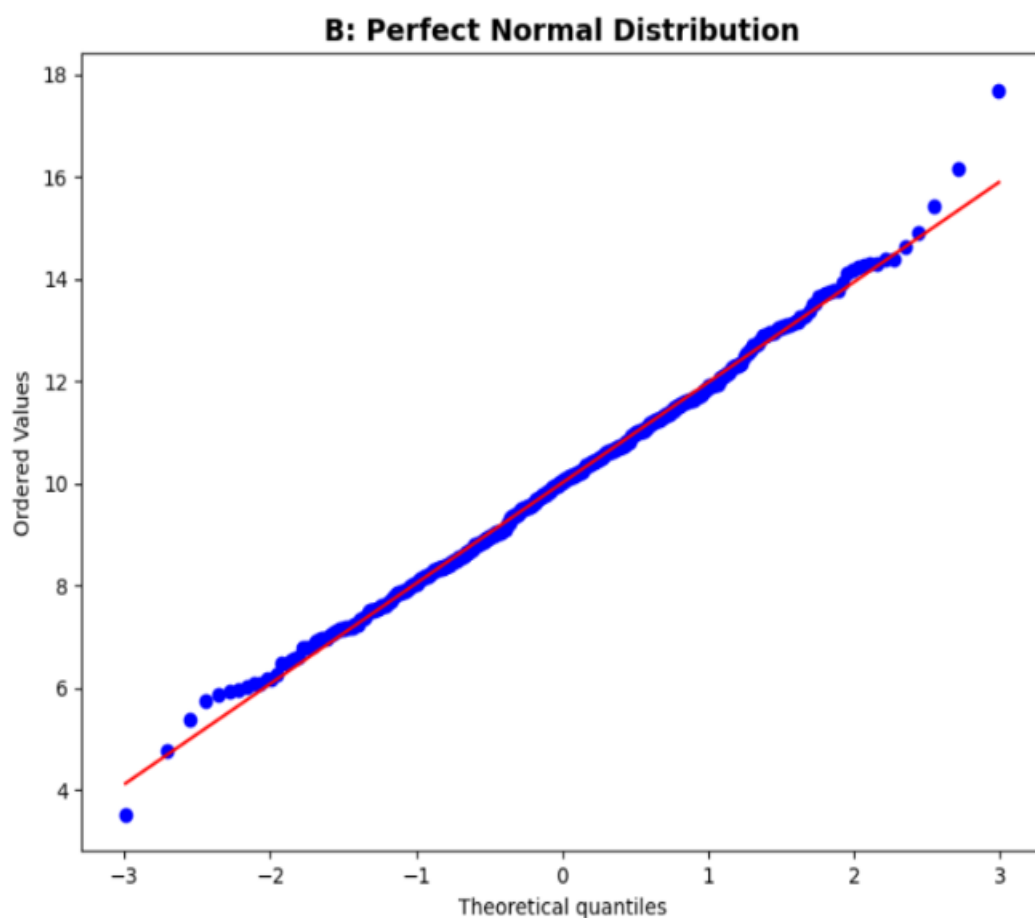
1. Normal Distribution (The Straight Line) :

- Visual: Points fall almost exactly along the 45-degree reference line.
- Significance: This represents a perfectly balanced population. If our data looked like this, it would mean that most mental disorders have a "medium" prevalence, with very few being extremely rare or extremely common.



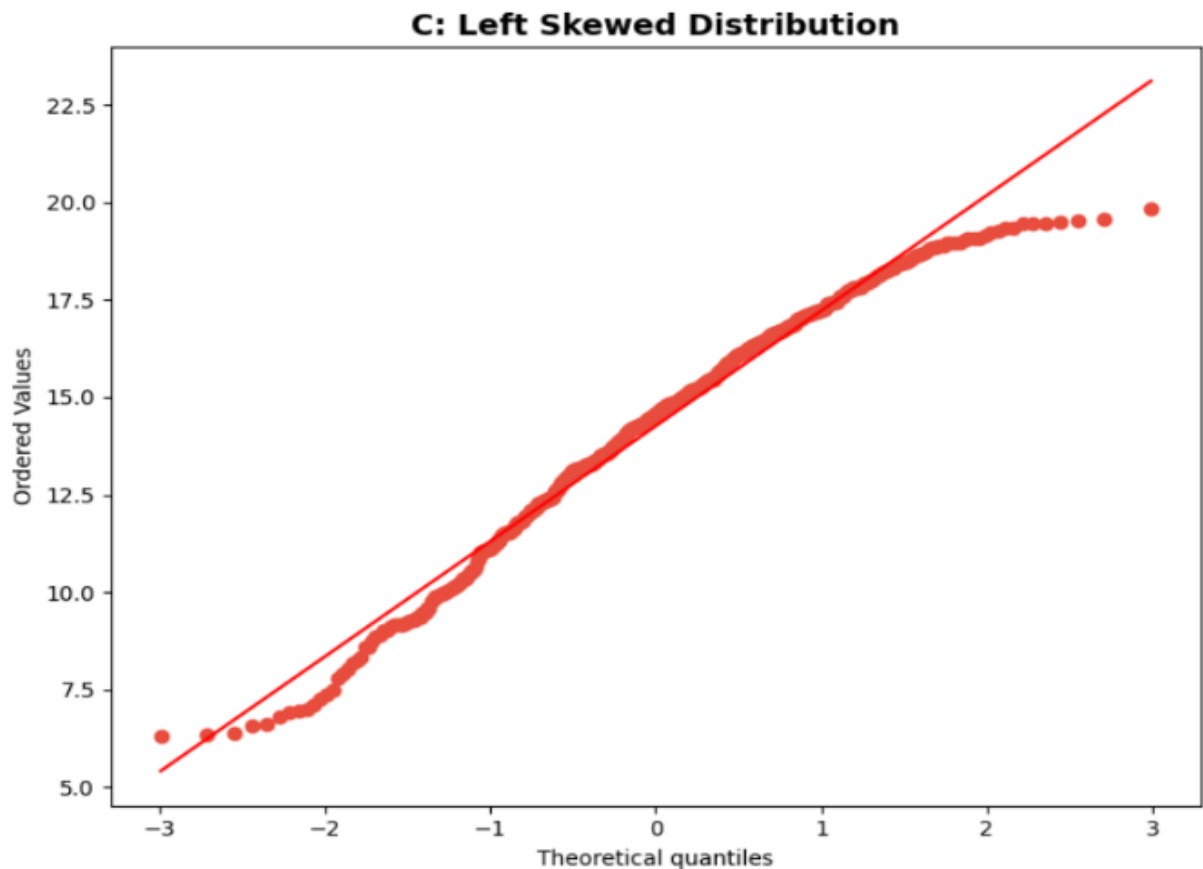
2. Right (Positive) Skewed (The Upward Curve)

- **Visual:** The points follow the line at first but curve **upward** away from the line at the high end (right side).
- **Significance:** This is the pattern found in our APMS data. It proves that while most disorders are infrequent, there are "extreme" high-prevalence categories (like Alcohol Use and Anxiety) that act as outliers.



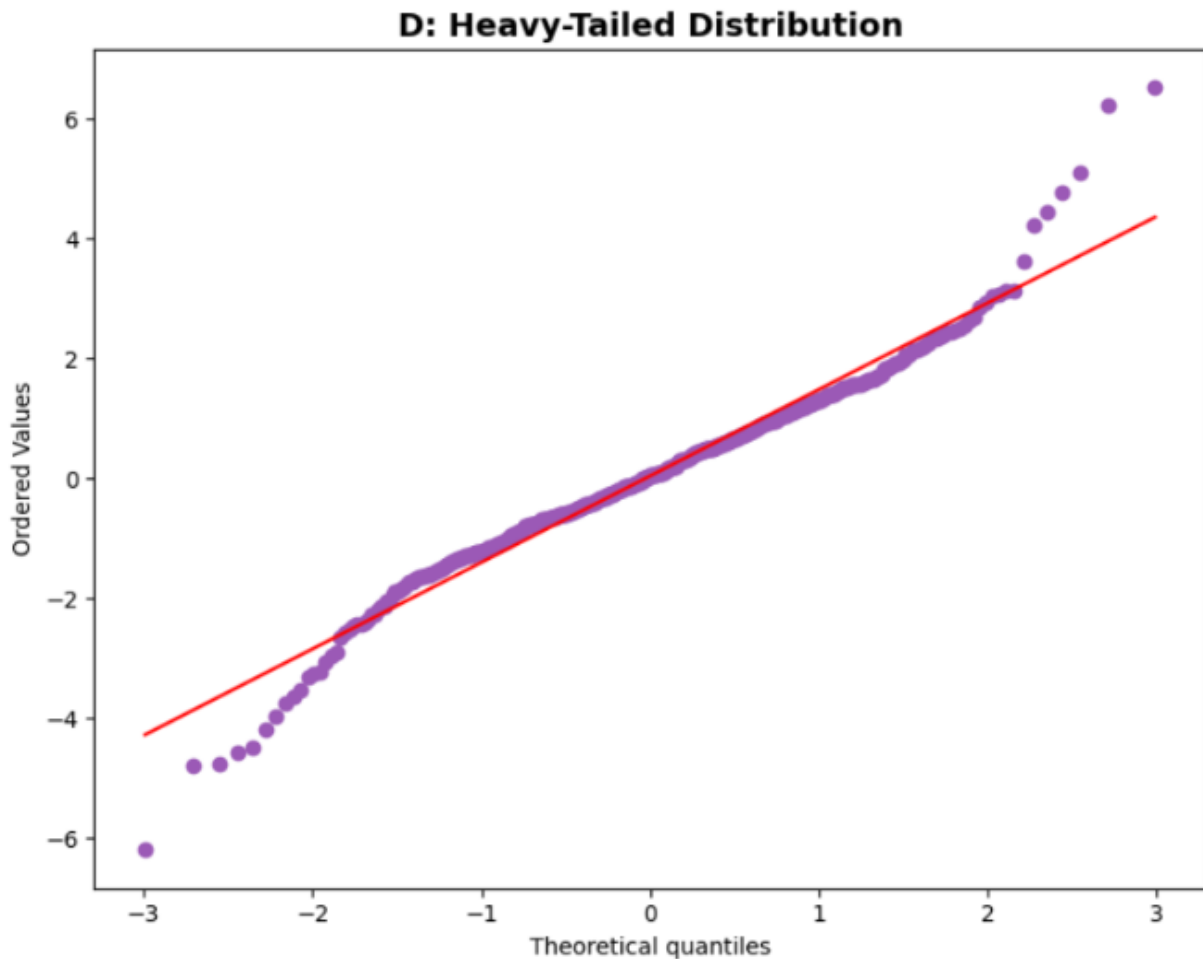
3. Left (Negative) Skewed (The Downward Curve)

- **Visual:** The points curve **downward** away from the line at the low end (left side).
- **Significance:** This would suggest a "ceiling effect," where most disorders are very common and only a few are rare. This is almost never seen in psychiatric prevalence data.



4. Heavy-Tailed (The "S" Curve) :

- Visual: Points curve below the line on the left and above the line on the right, forming an "S" shape.
- Significance: This indicates "Leptokurtic" data. It means the dataset has many more extreme outliers (both very high and very low) than a normal population would ever expect.

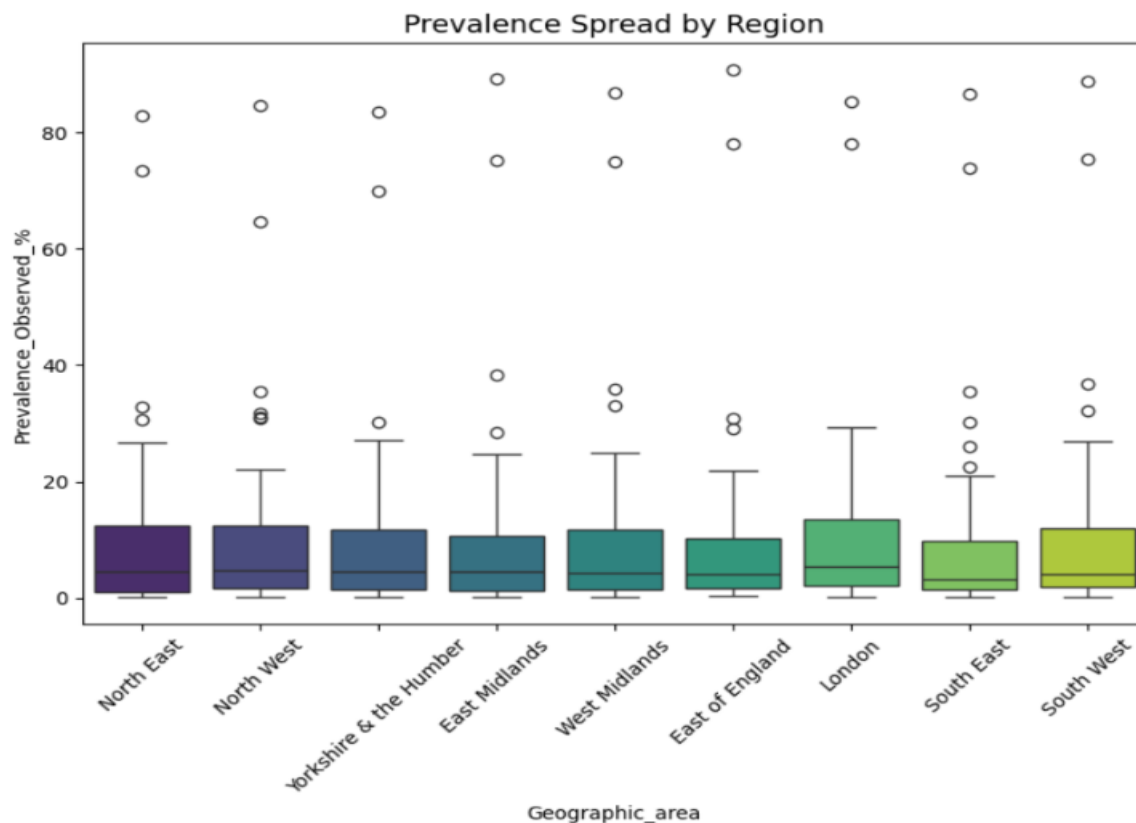


Q-Q (Quantile-Quantile) Plot Analysis

The Q-Q plot is used to determine if the prevalence rates follow a **Normal Distribution**.

- **Interpretation:** If the data were perfectly normal, all the blue dots would sit exactly on the red diagonal line.
- **The Curve:** In our analysis, the dots curve significantly away from the line at both ends. This "S-shape" or "U-shape" in Heavy-tailed graph indicates that the prevalence data has **heavy tails** and is skewed.
- **Analysis:** This confirms that the prevalence of mental disorders in the population is not "normally distributed".

3. Box Plot on Prevalence Spread by Region :



1. The Components

- **The Red Line:** This represents a perfect **Normal Distribution** (the "Bell Curve").
- **The Blue Dots:** These are our actual data points (the prevalence rates the dataset).
- **The X-Axis (Theoretical Quantiles):** Where the data *should* be if it followed a normal pattern.
- **The Y-Axis (Sample Quantiles):** Where the data *actually* is in our dataset.

2. The "S-Curve" Shape

The "S-shape" tells us :

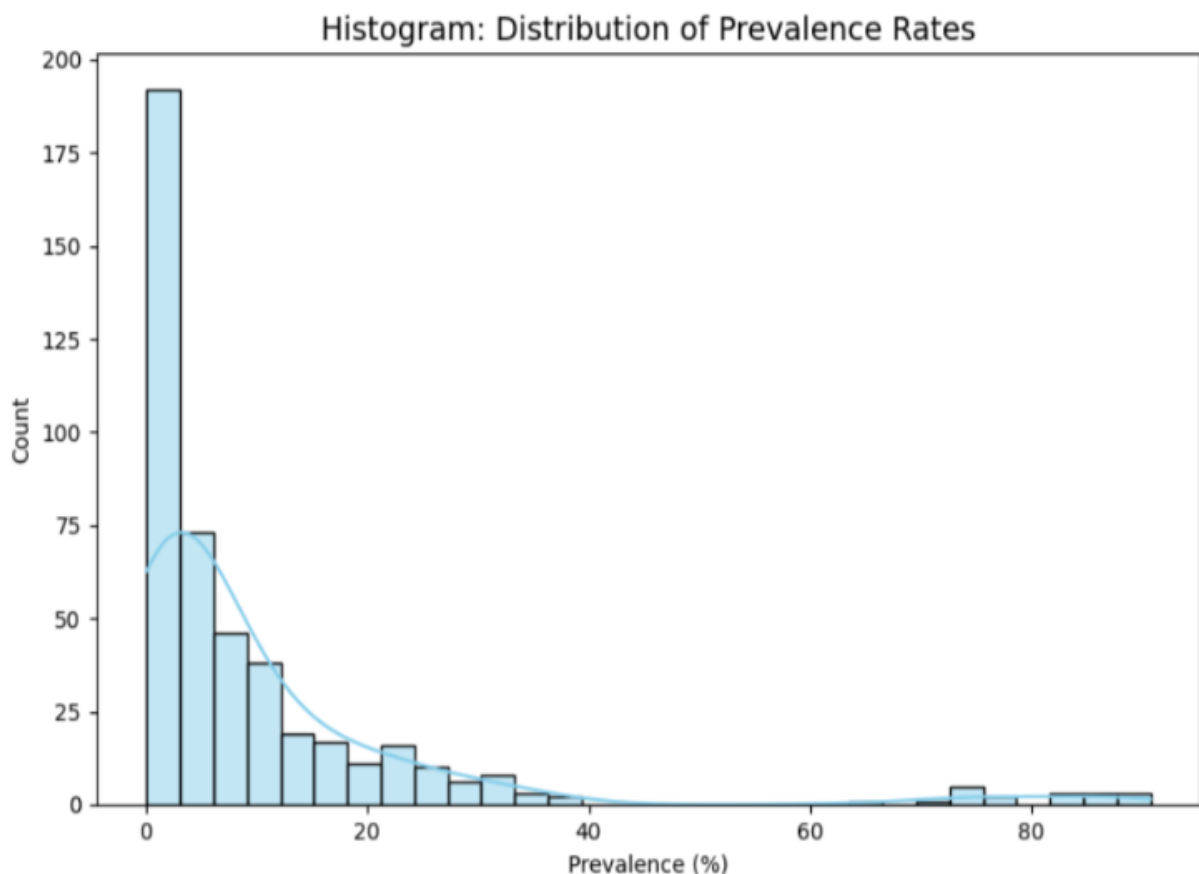
- **Heavy Right Tail:** The sharp curve upward on the right side means "extreme" high values (like Alcohol Use at 21%+) that are much higher than what a normal distribution would expect.

- **Non-Normality:** Because the dots don't stay on the red line, our data is **not normally distributed**. It is skewed.

3. Need of box plot :

- **The "Warning":** Since the dots drift off the line, a standard **T-Test** might be slightly less accurate because it assumes a perfect bell curve.
- **The "Solution":** This justifies why we also used the **Chi-Square test**. Chi-Square doesn't care if the data is a bell curve; it only cares about the raw counts, making it more reliable for this specific "S-curved" data.

4. Histogram Representation for Distribution of prevalence rates :



This chart is the final step of your **Distribution Analysis**. It compares the prevalence rates between **Men** and **Women** across all categories in the dataset.

5. Conclusion :

1. Evidence of Non-Normality

The visual evidence from the **Histogram** and **Q-Q Plots** consistently demonstrates that mental health prevalence does not follow a normal (bell-curve) distribution. Instead, the data is significantly **right-skewed**. This confirms that while the majority of psychiatric conditions affect a small percentage of the population, a few "heavy-use" categories (*specifically Common Mental Disorders and Alcohol Use*) exhibit much higher prevalence rates, pulling the population mean upward.

2. Validation of Statistical Methods

The identification of the "S-curve" and upward deviation in the Q-Q plots justifies the use of non-parametric tests (like the Chi-Square test) in earlier analysis. Because the data has "heavy tails" and high-prevalence outliers. The distribution analysis ensures that our findings regarding gender disparities are statistically robust.

3. Population Segments and Variance

The **segmented boxplots** revealed that while the median prevalence is relatively **stable** across *geographic regions*, the variance (spread) is significantly **higher** between **genders**.

This suggests that **gender** is a more powerful predictor of mental health distribution than geography within the 2014 UK context.